

Avneet Aujla

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SKILLS

Engineering & Design: AutoCAD, SolidWorks, Autodesk Inventor, ABAQUS, GD&T, Revit

Controls & Automation: TOYOPUC PLC, Allen-Bradley PLC, Ladder Logic, Structured Text (IEC 61131-3), Arduino

Machine Vision & Industrial Systems: Cognex, Machine Vision, Edge-Learning AI, Electronic Data Collection, Andon Systems

Programming & Software: Python, MATLAB, C, JavaScript, Linux, Microsoft Office Products

EXPERIENCE

Press Engineering Analyst Co-op, Toyota Motor Manufacturing Canada – Woodstock, ON Jan 2026 – Aug 2026

- Investigated recurring Nachi robot pneumatic valve failures causing **1.6 hours of downtime/month on a single robot**; implemented countermeasures **across 10 robots**, resulting in **zero repeat failures**.
- Developed and implemented a microscope based inspection method for RAV4 hood panels, enabling QC to **eliminate offline inspection of 100% of production panels** while maintaining upstream split detection.
- Designed in Solidworks and coordinated fabrication of a universal robot training dojo for **Woodstock and Cambridge auto palletizing systems**, enabling offline teaching of new model parts and robots, **eliminating production line teaching time**.

Project Engineering Intern, Cargill – Guelph, ON Apr 2025 – Aug 2025

- **Executed engineering deliverables across \$7M+ in capital projects**, spanning design, budgeting, corporate approvals, contractor coordination, and installation.
- Redesigned a livestock gate restraining system in AutoCAD, simplifying the mechanism to reduce fabrication cost while improving **operator safety, durability and usability**.
- Presented project scope and engineering solutions to **Cargill Protein North America Leadership** at corporate headquarters, communicating project requirements, costs, and implementation plans to senior stakeholders.
- Introduced a **standardized** engineering project tracking and documentation templates, **reducing approval and administrative turnaround time by 30%**.

PROJECTS

Automated Split Detection System, Toyota Motor Manufacturing Canada Aug 2026

- Developed an automated machine vision system after two tear-off events caused **35+ hours of die repair and 3 missed production runs**, creating risk of downstream production shortages.
- Designed and commissioned an in-press **Cognex** vision system integrating **Komatsu press and Nachi robot PLC** signals, angle based image triggering, HMI controls, and remote network monitoring.
- Trained and validated an edge learning model achieving **100% split detection with 0 false positives across 25+ test panels**, with PLC cycle stop integration for automatic line protection.

Electronic Data Collection Integration, Toyota Motor Manufacturing Canada Apr 2026

- Integrated **100+ Banner sensors** across a fully automated press line, configuring sensor data through PLC infrastructure to monitor **equipment vibration, temperature, and current** for predictive equipment health monitoring.
- Configured the Toyota EDC/Andon system to establish **equipment specific alarm thresholds and centralized fault data**, enabling maintenance teams to identify abnormal operating conditions and more quickly isolate potential failure points.
- Supported deployment and tuning of an **AI predictive maintenance system** that interprets equipment data and identifies the most likely failure cause; system was deployed on the production line with a **70% reduction in fault diagnostic time**.

Product Transfer Conveyor, Cargill Aug 2025

- Designed a custom product transfer conveyor in AutoCAD around existing spatial, safety, and material flow constraints, engineered to support **500+ pieces/hour**.
- Led design reviews and contractor coordination from concept through fabrication planning, modifying equipment interfaces to improve installation and maintenance feasibility.
- Improved material flow and reduced manual handling, achieving an estimated **95% reduction in production line backups**.

EDUCATION

University of Guelph | Guelph, ON | B.ENG. in Mechanical Engineering

Sept 2023 -